Exp. No: 4 Date:	Measurement and Analysis of Dispersion Compensation in Optical Fiber using OptiSystem

OBJECTIVES

1. To demonstrate dispersion compensation using the Ideal Dispersion Compensation component in OptiSystem.
2. To observe the broadening of a 12.5 ps optical Gaussian pulse after propagation through 10 km of single-mode fiber (SMF).
3. To compensate the accumulated dispersion of 160 ps/nm and verify recovery of the original pulse shape.

SOFTWARE / HARDWARE REQUIRED

- OptiSystem Software

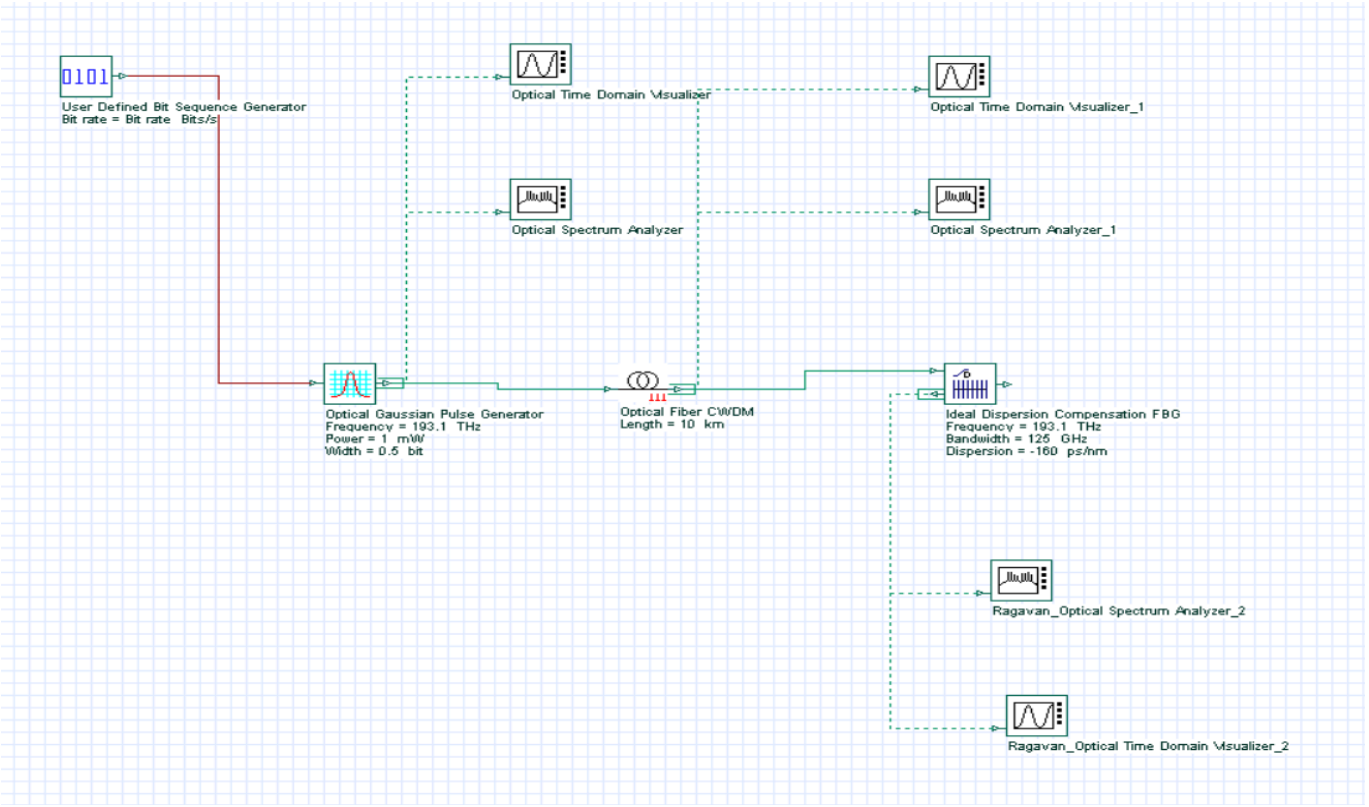
INPUT PARAMETERS

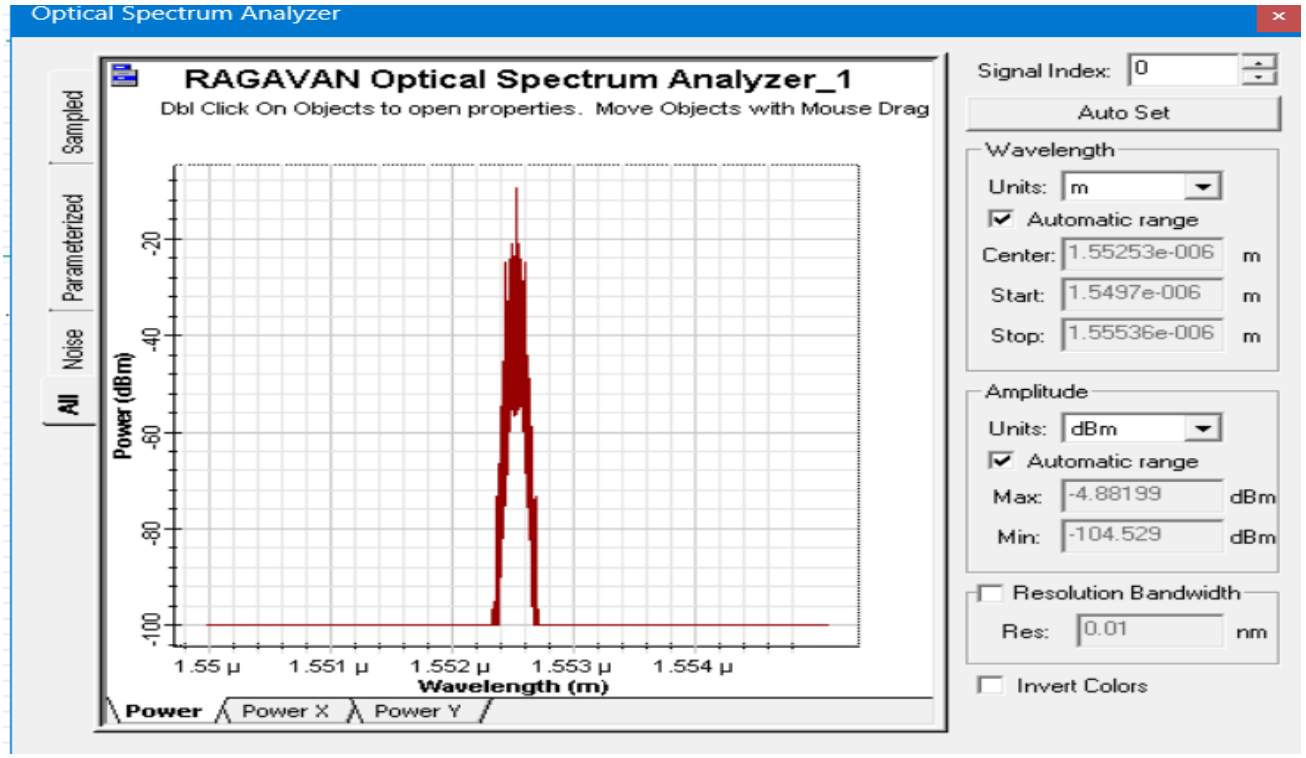
OptiSystem Block	Parameter	Value	Unit / Description
Global Parameters	Bit rate	40	Gb/s
Global Parameters	Pulse width / bit slot	0.5	× bit slot
Optical Gaussian Pulse Generator	Initial pulse width	12.5	ps
Optical Gaussian Pulse Generator	Frequency	193.1	THz
Optical Gaussian Pulse Generator	Power	1	mW
SMF	Length	10	km
SMF	Accumulated dispersion	160	ps/nm
Ideal Dispersion Compensation FBG	Frequency	193.1	THz
Ideal Dispersion Compensation FBG	Bandwidth	125	GHz (~3 × bit rate)
Ideal Dispersion Compensation FBG	Insertion loss	0	dB
Ideal Dispersion Compensation FBG	Depth	100	dB
Ideal Dispersion Compensation FBG	Dispersion	–160	ps/nm
Visualizers / Analyzers	Measurements	Pulse shape, spectrum	Performance analysis

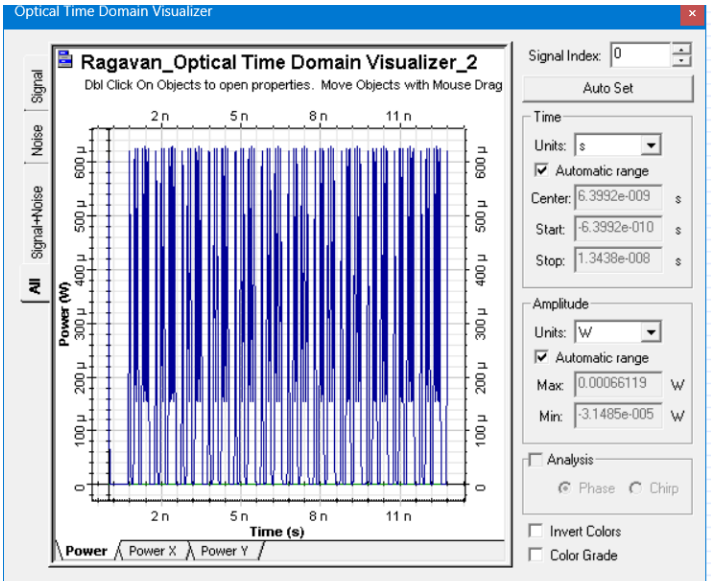
OUTPUT PARAMETERS

Stage	Observation Point	Result	Figure
Initial pulse	Gaussian pulse	Gaussian pulse, 12.5 ps	Figure 1
After propagation	Output of 10 km SMF	Pulse width increased ≈ 4 times	Figure 2
Compensator sejing	Ideal Dispersion Compensation FBG	Dispersion = –160 ps/nm	Figure 3
After compensation	Compensator output	Original pulse recovered (exact compensation)	Figure 4

OUTPUTS


0101
User Defined Bit Sequence Generator
Bit rate = Bit rate Bits/s
Optical Time Domain Visualizer
Optical Time Domain Visualizer_1
Optical Spectrum Analyzer
Optical Spectrum Analyzer_1
Optical Gaussian Pulse Generator
Frequency = 193.1 THz
Power = 1 mW
Width = 0.5 bit
Optical Fiber C\WDM
Length = 10 km
Ideal Dispersion Compensation FBG
Frequency = 193.1 THz
Bandwidth = 125 GHz
Dispersion = -160 ps/nm
Ragavan_Optical Spectrum Analyzer_2
Ragavan_Optical Time Domain Visualizer_2


Optical Spectrum Analyzer
RAGAVAN Optical Spectrum Analyzer_1
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Sampled
Parameterized
Noise
All
Power (dBm)
-20
-40
-60
-80
-100
1.55 µ
1.551 µ
1.552 µ
1.553 µ
1.554 µ
Wavelength (m)
Power
Power X
Power Y
Signal Index: 0
Auto Set
Wavelength
Units: m
Automatic range
Center: 1.55253e-006 m
Start: 1.5497e-006 m
Stop: 1.55536e-006 m
Amplitude
Units: dBm
Automatic range
Max: -4.88199 dBm
Min: -104.529 dBm
Resolution Bandwidth
Res: 0.01 nm
Invert Colors


Optical Time Domain Visualizer
Ragavan_Optical Time Domain Visualizer_2
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Signal
Noise
Signal+Noise
All
2 n
5 n
8 n
11 n
Power (W)
600 µ
500 µ
400 µ
300 µ
200 µ
100 µ
0
Time (s)
Power
Power X
Power Y
Signal Index: 0
Auto Set
Time
Units: s
Automatic range
Center: 6.3992e-009 s
Start: -6.3992e-010 s
Stop: 1.3438e-008 s
Amplitude
Units: W
Automatic range
Max: 0.00066119 W
Min: -3.1485e-005 W
Analysis
Phase
Chirp
Invert Colors
Color Grade

REFLECTION

I used OptiSystem to launch a 12.5 ps Gaussian pulse at 40 Gb/s through 10 km of single-mode fiber, where the accumulated dispersion of 160 ps/nm broadened the pulse to about four times its original width and would cause inter-symbol interference. I then set the Ideal Dispersion Compensation FBG to –160 ps/nm with a 125 GHz bandwidth centred at 193.1 THz, which restored the original pulse shape and showed that compensation is exact when the compensator value cancels the fiber's dispersion. I learned that the compensator's bandwidth and centre frequency matter as much as its dispersion value, since wrong settings leave the signal clipped or uncompensated. The component was ideal, with no loss or nonlinearity, so a real link would need to account for those effects.

LABORATORY PERFORMANCE ASSESSMENT RUBRIC

Objective	**Assessment Criterion**	**Excellent – 4**	**Good – 3**	**Satisfactory – 2**	**Needs Improvement – 1**	**Weightag e**
Design the OptiSystem layout	Project layout	Builds the complete layout with correct components and connections independently	Minor connection errors corrected	Needs guidance for components or connections	Unable to build the layout	20
Configure pulse, SMF and compensator parameters	Parameter configuration	Sets all parameters (40 Gb/s, 12.5 ps, 193.1 THz, 10 km, −160 ps/nm, 125	Minor parameter errors	Needs guidance for several parameters	Unable to configure parameters	20
Observe pulse broadening in SMF	Pulse observation	Captures and interprets initial and broadened pulses correctly	Minor interpretation errors	Limited observation / needs help	Unable to obtain or interpret pulses	20
Verify compensation using ideal dispersion component	Compensation verification	Verifies recovery of the pulse and explains residual effects of se:ings	Mostly correct verification	Partial verification with assistance	Unable to verify compensation	20
Document findings	Report and conclusion	Complete report with justified conclusion	Minor omissions	Incomplete record / weak conclusion	Report / conclusion inadequate	20

CONCLUSION

Dispersion compensation was successfully demonstrated using the Ideal Dispersion Compensation component in OptiSystem. A 12.5 ps Gaussian pulse broadened approximately four times after 10 km of SMF (accumulated dispersion 160 ps/nm), and sejing the compensator dispersion to –160 ps/nm restored the pulse, achieving an exact compensation of the accumulated dispersion.